

1. In your own words, what is the difference between a delegate and a lambda expression?

A delegate is like a type/contract for a method, while a lambda is a convenient way to provide the method's implementation.

delegates can be reused many times and make the program maintainable and easy to scale while delegates is a sugar syntax to write a simple method.

2. Why is var still statically typed even though the type isn't written explicitly?

Because it works and inferred the type of the variable in the compile time.

3. Give one real scenario where an anonymous type is more convenient than creating a full class.

trying to select some data inside a class or after some LINQ operations.

Anonymous types are useful for temporary data structures, especially when using LINQ to shape or project data.

4. When is ADO preferred to use over Entity Framework ?

- 1- in applications where performance is sensitive.
- 2- when I need all the control over the database.

5. Capacity Trick ?

is used to make the List data structure with a dynamic size and to optimize the time of adding from $O(n)$ to $O(1)$.

6. What is The Tuple ?

it is a data structure used to return a collection of items.